Forecasting and Explaining Inflation in a Data-Rich Environment: The Benefits of Machine Learning Methods

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Introduction

Inflation is one of the most important key economic indicators, which crucially determines the stability of the country's economy over a period of time. There has been a surge in the general price level recently after the COVID-19 pandemic and the Russian-Ukrainian war. This study aims to forecast the US inflation by comparing and evaluating classical time series models against the Machine Learning (ML) models. The analysis also identifies the key drivers of the US inflation prediction problem.

- A large monthly macroeconomic dataset including 783 observations and 119 variables, split into training data and low- and high-volatility test data.
- A comparison between benchmark models of Random Walk (RW), Autoregressive (AR), Generalized Autoregressive Score (GAS), AORW models, and ML models such as Ridge Regression (RR), LASSO, Adaptive LASSO (adaLASSO), Elastic Net (ElNet), Random Forest (RF), Extreme Gradient Boosting (XGBoost), and hybrid ML specifications.
- Identifying the most informative predictor groups and testing whether feature selection before estimation improves ML forecasting performance of the ML models.

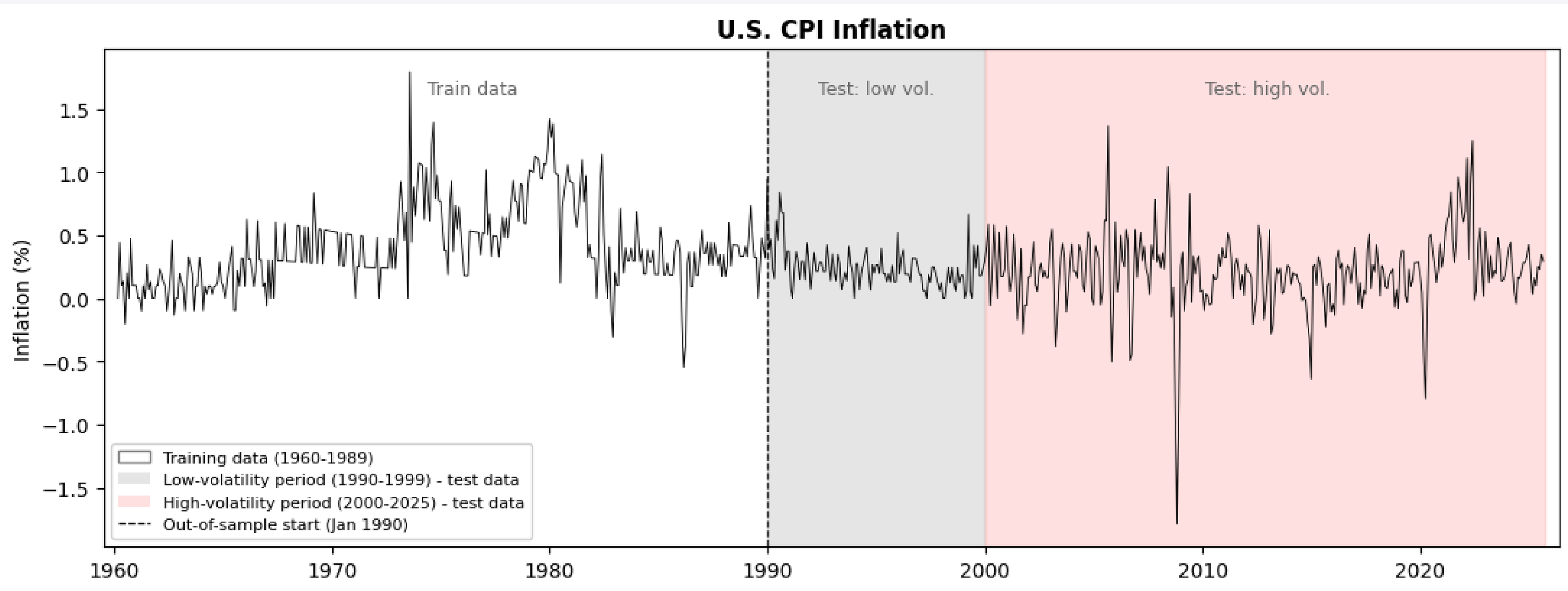

Figure 1: Monthly U.S. CPI Inflation

Methodology

To forecast the future US inflation, we consider a direct multi-step-ahead forecasting framework

$$\pi_{t+h} = F_h(\mathbf{X}_t) + u_{t+h}, \quad h = 1, 3, 6, 9, 12 \quad \& \quad t = 1, \dots, T$$

where π_{t+h} is inflation at month $t+h$ and u_{t+h} is a zero-mean random error, $\mathbf{X_t} = (x_{1t}, \dots, x_{nt})'$ is an $(n \text{ x } 1)$ input vector of predictors, and $F_h(\cdot)$ is the mapping function between predictors $\mathbf{X_t}$ and future inflation π_{t+h}. The direct forecasting equation form

$$\hat{\pi}_{t+h|t} = \hat{F}_{h,t-R_h+1:t}(\mathbf{X}_t).$$

where $\hat{F}_{h,t-R_h+1:t}$ is estimated target function using data from time $t - R_h + 1$ up to time t and R_h is the window size. This function depends on choice of model specification and forecasting horizon.

- **Benchmark Models**
  - **Random Walk:** is a popular basic model; the forecast is the last observed value of inflation
  - **AR(p):** a model depends on lagged values of inflation, and the lag order p is determined by BIC.
  - **AO-RW:** is a variant of the RW that produces a forecast of the average inflation over the last 12 months.
  - **GAS(1,1)**: an observation-driven model with a time-varying conditional mean of inflation, updated by the scaled score of the Gaussian likelihood, using an inverse-information scaling choice.
- **ML Models**.
  - **Ridge:** uses the L2 penalty term to shrink regression coefficients of unimportant variables close to zero.
  - **LASSO:** uses the L1 penalty term to shrink the coefficient of irrelevant variables exactly to zero.
  - **adaLASSO:** is an enhanced version of standard LASSO with an additional weighting parameter
  - **ElNet:** is a shrinkage model combines both L1 and L2 penalty terms, leading to an effective shrinkage.
  - **RF:** consists of multiple decision trees and takes the average of the individual trees' predictions.
  - **XGBoost:** is a decision tree-based ensemble model using a gradient boosting setup.
  - **Hybrid Models:** a shrinkage method (adapLASSO) is used to extract key features as input for ML models (RF, XGBoost) to make an inflation forecast.
- **Diebold-Mariano (DM) Test** is a statistical test used to compare the accuracy of two forecasting models. This is done by comparing the forecast errors of two models. This test calculates a DM test statistic that measures the standardized loss differential of two models (both square loss and absolute loss are considered in the analysis).

Results: Forecasting Performance

The average of the root mean squared error (RMSE), mean absolute error (MAE), and median absolute deviation from the median (MAD), relative to the RW model across all forecasting horizons.

	Forecasting precision					
	Less Volatile (1990–1999)			More Volatile (2000–2025)		
Model	Aver RMSE	Aver MAE	Aver MAD	Aver RMSE	Aver MAE	Aver MAD
RW	1.000	1.000	1.000	1.000	1.000	1.000
AR(p)	0.874	0.956	0.719	0.796	0.764	0.749
AORW	**0.781**	**0.796**	0.729	0.791	0.783	0.756
GAS(1,1)	0.867	0.957	0.699	0.789	0.751	0.719
Ridge	0.872	0.952	0.795	0.784	0.755	0.752
LASSO	0.858	0.922	0.792	0.794	0.759	0.742
adaLASSO	0.866	0.935	0.771	0.781	0.750	**0.702**
Elastic Net	0.867	0.936	0.782	0.794	0.758	0.724
RF	0.797	0.865	**0.640**	**0.746**	**0.720**	**0.702**
XGBoost	0.853	0.896	0.773	0.808	0.795	0.764
adaLASSO-RF	0.816	0.880	0.705	0.792	0.764	0.721
adaLASSO-XGB	0.866	0.924	0.818	0.848	0.836	0.804

Table 1: Average RMSE, MAE, and MAD across all horizons, for the less-volatile and more-volatile subsamples, relative to the RW model. The average metrics for the best-performing models are highlighted in bold.

Discussion: Forecasting Performance

- Most of the models have worse performance during low-volatility periods than high-volatility ones in relative to RW.
- AORW is the best-performing model in terms of less-volatile RMSE and MAE.
- Random Forest outperforms other models, achieving the lowest prediction error, especially within more volatile periods.
- Tree-based models with all variables perform better than their hybrid specifications.
- Variable selection appears to be less useful for improving the model performance.
- The DM test results also confirm that RF produces a more accurate forecast than each of the benchmark models.

Compared with the literature:

- The obtained results align with the study [1], where RF is still the dominant model for US CPI inflation forecasting, even with an updated U.S. dataset and a different model set.
- The findings, however, seem not to be universal in other countries' CPI inflation forecasts, such as China [2], Turkey [3], since tree-based models like Random Forest do not have superior performance, and feature selection supports the model's predictive performance.
- This difference in the results and findings varies due to the nature of each country's economy, different inflation dynamics, and the predictor set.

Results: Explaining

Figures 2 and 3 show histograms and line charts for the relative importance of each group variable for each forecasting horizon on the full and two subsample periods for the **Random Forest** model.

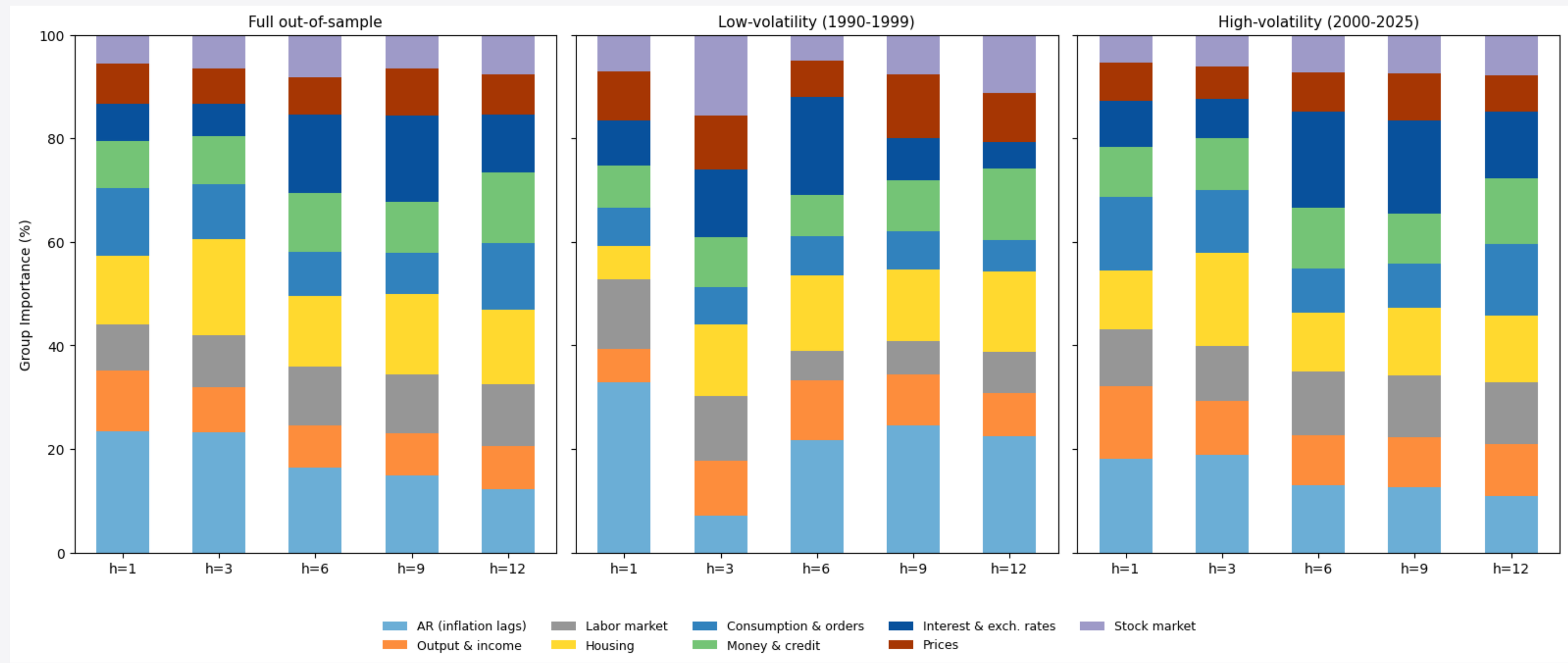

Figure 2: Histogram for Variable Group Importance for US CPI Inflation

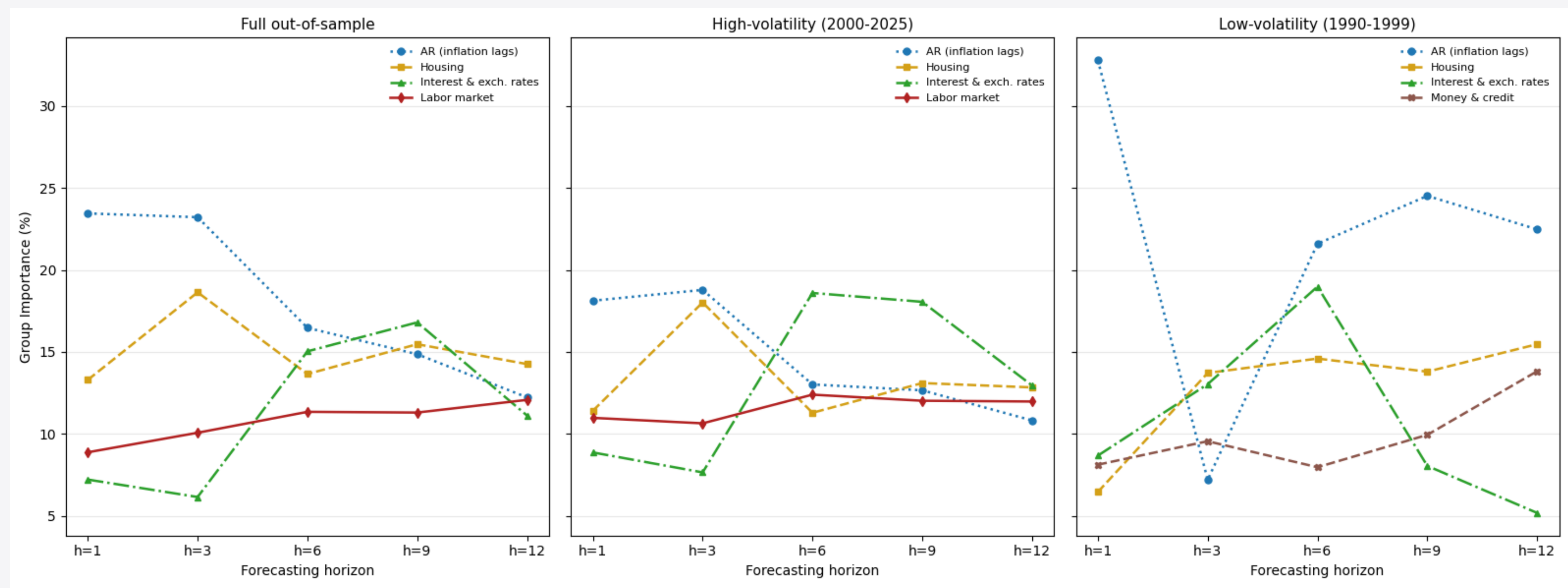

Figure 3: Four Main Variable Groups Importance Across Forecasting Horizons for US CPI Inflation

- The top four most important groups are the same for the full and high-volatility sample periods.
- The dominance of inflation lags implies that US CPI inflation is strongly persistent.
- The importance of lagged inflation declines for longer forecasting horizons, while housing and interest & exchange rates gain their weights for the medium and longer horizons.
- AR terms, Housing, Interest exchange rates, and Labor market exhibit similar evolutionary patterns across the full and high-volatility sample periods.
- In less-volatile periods, the labor market drops out of the top four, replaced by Money & credit.

Conclusion

- ML models applied to a large dataset outperform traditional time-series benchmarks, particularly in volatile periods, confirming the benefits of ML methods within a data-rich environment for US CPI inflation forecasting.
- Random Forest is the dominant model in the US CPI inflation forecasts, and it helps identify the key drivers of US CPI inflation dynamics.
- Feature selection does not improve and often slightly worsens the ML forecasting performance.

References

[1] Medeiros, M.C., Vasconcelos, G.F.R., Veiga, Á. & Zilberman, E. (2021). Forecasting inflation in a data-rich environment: the benefits of machine learning methods. *Journal of Business & Economic Statistics*, 39(1), 98–119.

[2]Huang, N., Qi, Y. and Xia, J., 2025. China's inflation forecasting in a data-rich environment: based on machine learning algorithms. Applied Economics, 57(17), pp.1995-2020.

[3] Aras, S. & Lisboa, P.J.G. (2022). Explainable inflation forecasts by machine learning models. *Expert Systems with Applications*, 207, 117982.

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